vRadio

Yaesu FTDX101D as a Kenwood TS-2000 for N1MM Logger+ | manual, 10 Nov 2026

vRadio connects to the FTDX101D on **COM14 (38400 8N1)** and presents the radio's **VFO-B (SUB)** to N1MM Logger+ as **radio 2**, a Kenwood TS-2000, over **TCP port 4532**. N1MM radio 1 stays on its own connection to the radio's VFO-A (MAIN).

```
FTDX101D --- COM14 38400 8N1 ---> vRadio.exe --- TCP 127.0.0.1:4532 ---> N1MM radio 2 (TS-2000)
                                        ^
                                        +--- UDP 12060 <--- N1MM RadioInfo broadcast (TX focus)
```

Status: everything confirmed on air except the dual RX value (see section 8).

1. Start vRadio

Double-click `dist\vRadio_TEST.bat`, or run the full working command (with debug log):

```
C:\Ham\vRadio\dist\vRadio.exe --tcp 4532 --follow-band --auto-split --n1mm-udp 12060
    --dual-rx --dual-rx-cmd FR00; -v --log C:\Ham\vRadio\dist\vradio_test_log.txt
```

Normal use, without the debug log:

```
C:\Ham\vRadio\dist\vRadio.exe --tcp 4532 --follow-band --auto-split --n1mm-udp 12060
    --dual-rx --dual-rx-cmd FR00;
```

Without dual RX (if `FR00;` turns out to be wrong):

```
C:\Ham\vRadio\dist\vRadio.exe --tcp 4532 --follow-band --auto-split --n1mm-udp 12060
```

System tray version

Double-click `dist\vRadio Tray.lnk` (runs `dist\vRadioTray.exe` with the normal-use options above). It has no console window and sits in the system tray next to the clock. Start with Windows: copy the shortcut into the Startup folder (Win+R, `shell:startup`).

Icon	Meaning
green	radio answers
red	radio not answering, or an error (hover or right-click for details)
orange	transmitting
grey	starting / connecting

Right-click menu: radio status, N1MM connection, TX focus, *Copy VFO-A -> VFO-B (same band)*, *Open log*, *Open vRadio folder*, *Exit vRadio*. Log: `dist\vradio_log.txt` (rotating, max 4 x 2 MB). Only one vRadioTray can run at a time. Same options as vRadio.exe.

Radio on another port: put the port and speed first, e.g.:

```
C:\Ham\vRadio\dist\vRadio.exe COM5 38400 --tcp 4532 --follow-band --auto-split --n1mm-udp 12060
```

Stop: close the vRadio window or press Ctrl+C. Only one program can use COM14 at a time.

2. N1MM setup

Where	Setting
Configurer > Hardware, radio 2	Port **TCP**, Radio **Kenwood TS-2000**, IP **127.0.0.1**, port **4532**
Configurer > Broadcast Data	**Radio** ticked, address **127.0.0.1:12060** (vRadio reads the TX focus from it)

3. How it works

Same band for both radios

N1MM refuses to put radio 1 and radio 2 on one band ("Can not set both radios to the same band!"), but accepts it when the radio does it. With `--follow-band`, every time VFO-A changes band vRadio sends `AB;` (copy VFO-A to VFO-B, frequency and mode). You can still tune radio 2 anywhere afterwards.

Auto split by N1MM TX focus

- N1MM keys most transmissions (WinKeyer CW) without CAT, so vRadio follows N1MM's **TX focus** (`ActiveRadioNr` in the RadioInfo UDP broadcast).
- Radio 2 has TX focus: vRadio sends `ST1;` - TX on VFO-B.
- Radio 1 has TX focus: vRadio sends `ST0;` - split off, TX on VFO-A.
- A focus change during a transmission waits until the radio is back on RX. After every TX-to-RX the TX VFO is set again.
- N1MM sees radio 2 as a plain, non-split radio.

Dual RX

With `--dual-rx`, vRadio reads `FR;` every 2 s and re-sends `--dual-rx-cmd` (`FR00;`) if the radio reports something else. This overrides manual dual-RX toggles.

4. All options

Option	Effect
`COM5 38400` (first, no dashes)	Radio port and speed, e.g. `vRadio.exe COM5 38400 --tcp 4532`. Same as --radio / --radio-baud. Default COM14 38400.
`--tcp 4532`	Serve the TS-2000 emulation on TCP 4532 instead of a COM port.
`--tcp-host 0.0.0.0`	Accept TCP from other PCs (default 127.0.0.1 = this PC only). Allow vRadio.exe in Windows Firewall.
`--vfo B`	Default. N1MM VFO-A = radio VFO-B (SUB), N1MM VFO-B = radio VFO-A. `--vfo A` = normal mapping.
`--follow-band`	When radio VFO-A changes band, VFO-B jumps to that band (`AB;`).
`--same-band`	Copy VFO-A to VFO-B once at startup.
`--auto-split`	vRadio sets the TX VFO itself (`ST1;` = VFO-B, `ST0;` = VFO-A). CAT `TX;` from radio 2 transmits on VFO-B.
`--n1mm-udp 12060`	With --auto-split: follow N1MM TX focus from the RadioInfo broadcast.
`--n1mm-radio 2`	Which N1MM radio number vRadio is (default 2).
`--dual-rx`	Keep dual receive on.
`--dual-rx-cmd FR00;`	Command for dual receive (default FR00;, verify on site).
`--tx-follow`	At startup set TX to the controlled VFO. Ignored with --auto-split.
`--usb-data / --lsb-data`	USB/LSB from the software selects DATA-U/DATA-L (FT8 etc.).
`--radio COM14 --radio-baud 38400`	Radio port and speed (defaults).
`--cat COM1 --cat-baud 38400`	COM-port mode instead of TCP (software uses COM2 of the VSPE pair).
`-v / --log FILE`	Log every command / also write the log to a file (rotating, 4 x 2 MB).
`--tray`	Run as a tray icon (vRadioTray.exe always does).

N1MM macro: `{CAT2ASC VRAB;}` copies VFO-A to VFO-B on a key press.

5. Files

File	Purpose
`dist\vRadio.exe`	The translator, console version (no Python needed).
`dist\vRadioTray.exe`	The same, as a system tray app (no console).
`dist\vRadio Tray.lnk`	Shortcut: vRadioTray with the full working options.
`dist\vRadio_TEST.bat`	Full working setup with debug log.
`dist\vRadio_N1MM_TCP.bat`	Older launcher: TCP + follow-band only (no auto-split).
`dist\radio_monitor.exe / .bat`	Read-only monitor of FR/FT/ST/VS/TX/FA/FB/MD; prints every change. Stop vRadio first.
`dist\update_vRadio.bat`	Copies dist\new\vRadio.exe over dist\vRadio.exe after a rebuild.
`vradio.py`	Source code.
`probe_yaesu.py, test_client.py, test_tcp.py`	Small test tools (Python).
`README.md, NOTES_auto_split.md`	Documentation and background notes.
`make_manual_pdf.py`	Builds this PDF.

6. Rebuild the exe

```
cd C:\Ham\vRadio
"%LOCALAPPDATA%\Programs\Python\Python312\python.exe" -m PyInstaller
    --onefile --console --icon vradio.ico --name vRadio vradio.py
"%LOCALAPPDATA%\Programs\Python\Python312\python.exe" -m PyInstaller
    --onefile --windowed --icon vradio.ico --name vRadioTray vradio.py
```

(Two commands, each written on two lines here.) Then rebuild this manual with `python make_manual_pdf.py`.

Python 3.12, pyserial and PyInstaller are installed for this Windows user.

7. TS-2000 commands supported

ID, TY, AI (emulated by polling), PS, FA, FB, IF, MD, TX, RX, FT, FR, VV, SM (0-255 to 0-30), AG, RG, SQ, PC, KS, VX, NB, NR, PA, RA, RT, XT, RC, RU, RD, BY, UP, DN, plus vRadio's own VRAB. Anything else is answered `?;`. Not mapped: KY (CW text), SH/SL (filters), EX (menu).

8. Troubleshooting and open items

Problem	What to do
COM14 busy	A vRadio or radio_monitor is still running. Close it, or end vRadio.exe in Task Manager (the one-file exe runs as two processes; end both).
Radio does not answer (no ID0681;)	COM14 is an ELTIMA virtual port, not the radio's USB. Start the program that links COM14 to the radio; check the radio is on.
"Can not set both radios to the same band!"	Use --follow-band or the VRAB macro.
TX on the wrong VFO	Check N1MM Broadcast Data > Radio > 127.0.0.1:12060 and look for "N1MM TX focus" lines in the log.

Open item - confirm dual RX on site:

- Close vRadio.
- Run `C:\Ham\vRadio\dist\radio_monitor.exe COM14 38400`.
- Toggle dual RX (Ctrl+Alt+D in N1MM or the radio button) and note the `FR;` value with both sides receiving.
- If it is not `FR00;`, use that value after `--dual-rx-cmd`.